

Board of Studies (BoS) Syllabus Template

Course Information

Degree: UG

Program: BCA

Course Name: .NET PROGRAMMING

Course Credits: 03

Semester: III

Course Category: Elective (DSE)

Type of Course: Lecture

New Course: No

Learning Hours

Activity	Hours per Week
Lecture	2
Tutorial	0
Practical	2
Experiential Learning	0

Course Contribution

Category	Percentage Contribution
Employment Enhancement	40%
Entrepreneurship Skills	30%
Skill Development	30%

Course Relevance

Level	Percentage Relevance
Global	40%
National	30%
Regional	20%

Local	10%
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Course Objectives (COB)

CO#	Course Objectives
C01	Understand the foundations of CLR execution.
C02	Learn the technologies of the .NET framework.
C03	Know the object oriented aspects of C#.
C04	Be aware of application development in .NET.
C05	Learn web based applications on .NET (ASP.NET).

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
C01	List the major elements of the .NET frame work	L2
C02	Explain how C# fits into the .NET platform.	L2
C03	Analyze the basic structure of a C# application	L3
C04	Develop programs using C# on .NET	L4
C05	Design and develop Web based applications on .NET Discuss CLR	L6

SESSION PLAN :

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percent age change in Module (put Nil if there is no change or it is a new course write NEW)
MODULE 1 : INTRODUCTION TO C#	Introducing C#, Understanding .NET	PPT Presentation / Classroom Interaction	BL-2	CO2, CO1	NEW
	overview of C#	PPT presentation	BL-3	CO2	NEW
	Literals, Variables, Data Types	PPT presentation	BL-1	CO3	NEW
	Operators, checked and unchecked operators	PPT presentation	BL-3	CO3	NEW
	Expressions, Branching, Looping	Hands on activity	BL-3	CO3, CO4	NEW
	Methods, implicit and explicit casting, Constant	Hands on activity	BL-3	CO4, CO3	NEW
	Arrays, Array Class, Array List	Hands on activity	BL-3	CO4	NEW
	String, String Builder, Structure	PPT presentation	BL-3	CO4	NEW
	Enumerations	PPT presentation	BL-3	CO4	NEW
	boxing and unboxing	Hands on activity	BL-4	CO3	NEW
MODULE 2 : OBJECT ORIENTED ASPECTS OF C#	Class	PPT presentation	BL-2	CO2	NEW
	Objects	PPT presentation	BL-3	CO2	NEW
	Constructors and its types	PPT presentation	BL-3	CO2	NEW

	inheritance, properties	Hands on activity	BL-3	C02	NEW
	indexers, index overloading	Live Demo	BL-3	C02	NEW
	polymorphism	PPT presentation	BL-4	C02	NEW
	sealed class and methods	Hands on activity	BL-2	C02	NEW
	interface	PPT presentation	BL-3	C02	NEW
	abstract class, abstract and interface	Hands on activity	BL-4	C02	NEW
	operator overloading	PPT presentation	BL-3	C02	NEW
	delegates, events	Live Demo	BL-3	C03	NEW
	errors and exception, Threading	Live Demo	BL-4	C03	NEW
MODULE 3 : APPLICATION DEVELOPMENT ON .NET	Building windows application	PPT presentation	BL-3	C03	NEW
	Creating our own window forms with events and controls	PPT presentation	BL-3	C03	NEW
	menu creation, inheriting window forms	PPT presentation	BL-3	C03	NEW
	SDI and MDI application	Live Demo	BL-3	C03	NEW
	Dialog Box(Modal and Modeless), accessing data with ADO.NET	Hands on activity	BL-3	C03	NEW
	DataSet, typed dataset, Data Adapter	Hands on activity	BL-3	C03	NEW
	updating database using stored procedures	PPT presentation	BL-3	C03	NEW
	SQL Server with ADO.NET	PPT presentation	BL-3	C03	NEW
	handling exceptions, validating controls	Hands on activity	BL-4	C03	NEW
	windows application configuration	Hands on activity	BL-5	C03	NEW

MODULE 4 : WEB BASED APPLICATION DEVELOPMENT ON .NET	Programming web application with web forms	PPT presentation	BL-3	C04	NEW
	ASP.NET introduction	PPT presentation	BL-2	C01	NEW
	working with XML and .NET	Live Demo	BL-3	C04	NEW
	Creating Virtual Directory and Web Application	Live Demo	BL-3	C04	NEW
	session management techniques, web.config	Hands on activity	BL-3	C04	NEW
	web services, passing datasets	Live Demo	BL-3	C04	NEW
	returning datasets from web services	Hands on activity	BL-3	C04	NEW
	handling transaction, handling exceptions	Live Demo	BL-4	C04	NEW
	returning exceptions from SQL Server	Live Demo	BL-4	C04	NEW
MODULE 5 : CLR AND .NET FRAMEWORK	Assemblies	PPT presentation	BL-2	C01	NEW
	Versioning	PPT presentation	BL-2	C01	NEW
	Attributes	PPT presentation	BL-3	C02	NEW
	reflection	Hands on activity	BL-4	C02	NEW
	viewing meta data	Live Demo	BL-3	C02	NEW
	type discovery	Hands on activity	BL-3	C02	NEW
	reflection on type	Live Demo	BL-4	C02	NEW
	marshalling	Live Demo	BL-3	C03	NEW
	remoting	Live Demo	BL-3	C03	NEW
	security in .NET	Hands on activity	BL-5	C05	NEW

